Day_1_Lecturer_Notes_Linear_Ridge_Regression

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1 Lecturer Notes: Linear & Ridge Regression Analysis for Thermofluid Systems

These notes are to explain the foundations of:

1. **Linear Regression**, specifically Ordinary Least Squares (OLS)
2. **Ridge Regression**, also known as L2-regularised regression
3. Model assessment using **MAE**, **RMSE**, and R^2

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2 Linear Regression: Ordinary Least Squares

Linear regression models the relationship between a dependent variable, usually denoted y, and one or more independent variables, usually denoted X.

In a thermofluid context:

- y might be a heat transfer coefficient, pressure drop, wall temperature, Nusselt number, drag coefficient, or compressor efficiency.
- X might include Reynolds number, Prandtl number, Mach number, flow velocity, pressure, density, surface roughness, inlet temperature, or sensor readings.

Linear regression assumes that the response can be approximated as a **linear combination of the input features**.

2.1 The Linear Regression Model

For one observation, the model is:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \epsilon$$

where:

- β_0 is the intercept.
- $\beta_1, \beta_2, \ldots, \beta_p$ are the regression coefficients.
- $x_1, x_2, \ldots, x_p$ are the input features.
- ϵ is the error term, representing measurement noise, unmodelled physics, or random uncertainty.
- p is the number of input features.

For a dataset with n observations, the model can be written in matrix form as:

$$y = X\beta + \epsilon$$

where:

- y is an $n \times 1$ target vector.
- X is an $n \times (p+1)$ design matrix if an intercept column is included.
- β is a $(p+1) \times 1$ coefficient vector.
- ϵ is an $n \times 1$ error vector.

If the model includes an intercept, the first column of X is usually a column of ones.

2.2 The OLS Objective Function

Ordinary Least Squares chooses the coefficient vector $\hat{\beta}$ that minimises the **residual sum of squares**.

This quantity is best called **RSS** or **SSE**:

- **RSS**: residual sum of squares
- **SSE**: sum of squared errors

Avoid using **SSR** for this quantity unless you define it carefully, because in many statistics texts **SSR** means **regression sum of squares**, which refers to explained variation rather than residual error.

The residual sum of squares is:

$$\text{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

OLS solves:

$$\min_{\beta} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

or equivalently:

$$\min_{\beta} \|y - X\beta\|_2^2$$

where:

- y_i is the measured or observed value.
- $\hat{y}_i$ is the model prediction.
- $y_i - \hat{y}_i$ is the residual.

2.3 Analytical Solution: The Normal Equation

If $X^T X$ is invertible, the OLS solution is:

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

This is often called the **normal equation**.

However, this formula requires the columns of X to be linearly independent. If $X^T X$ is singular or nearly singular, the inverse may not exist or may be numerically unstable.

In practice, numerical methods such as QR decomposition, singular value decomposition, or the Moore-Penrose pseudoinverse are often used instead of explicitly computing $(X^T X)^{-1}$.

2.4 When OLS Works Best

OLS works best when:

- the relationship is approximately **linear in the model parameters**,
- residuals are approximately independent,
- residual variance is roughly constant, known as **homoscedasticity**,
- multicollinearity among predictors is not severe,
- the dataset does not contain extreme influential outliers.

Normally distributed residuals are **not required** to compute the OLS fit. However, approximate normality is useful for classical statistical inference, such as confidence intervals, prediction intervals, and hypothesis tests.

2.5 Important Limitations of OLS

2.5.1 Sensitivity to Outliers

OLS squares the residuals. This means large errors receive disproportionately large penalties.

In thermofluid experiments, an outlier might come from:

- a faulty pressure transducer,
- a thermocouple spike,
- a flow-rate logging error,
- a failed CFD run,
- transient behaviour accidentally included in steady-state data.

A single severe outlier can significantly affect the fitted coefficients.

2.5.2 Multicollinearity

Multicollinearity occurs when two or more predictors are highly correlated.

In thermofluid systems, this is common because many variables are physically coupled. For example:

- pressure, density, and temperature may be related through an equation of state,
- Reynolds number depends on velocity, density, viscosity, and length scale,
- Nusselt number and heat transfer coefficient are directly related,
- neighbouring sensors may measure similar thermal or pressure behaviour.

When predictors are highly collinear, $X^T X$ can become nearly singular. This makes coefficient estimates unstable. Small changes in the data can lead to large changes in the fitted coefficients.

This does not always destroy prediction accuracy, but it can make coefficient interpretation unreliable.

2.5.3 Poor Extrapolation

A linear model may fit well inside the observed experimental range but perform poorly outside it.

This is especially important in thermofluid systems because the underlying physics may change between regimes, for example:

- laminar to transitional flow,
- transitional to turbulent flow,
- subsonic to compressible flow,
- single-phase to two-phase flow,
- forced convection to mixed or natural convection.

A regression model should not be trusted far outside the domain of the data used to train it unless there is strong physical justification.

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3 Ridge Regression: L2 Regularisation

In many real thermofluid datasets, variables are strongly coupled. Including many correlated measurements in an OLS model can lead to unstable coefficients and poor generalisation.

Ridge Regression addresses this by adding a penalty on the size of the coefficients.

Ridge regression is also called **L2-regularised linear regression**.

3.1 Ridge Objective Function

Ridge regression minimises:

$$\min_{\beta} \left[\sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right]$$

where:

- λ is the regularisation strength.
- λ is often called α in machine-learning software.
- $\sum_{j=1}^{p} \beta_j^2$ is the **L2 penalty**.
- p is the number of input features.

The intercept is usually **not penalised**. The penalty is normally applied only to the feature coefficients, not to the constant offset term.

3.2 Ridge Solution

If all coefficients are penalised and the intercept is handled separately, the ridge solution is commonly written as:

$$\hat{\beta}_{ridge} = (X^T X + \lambda I)^{-1} X^T y$$

where:

- I is the identity matrix.
- λI adds a positive value to the diagonal of $X^T X$.
- $\lambda > 0$ controls the strength of the penalty.

A more precise version, when the intercept is included in X but not penalised, is:

$$\hat{\beta}_{ridge} = (X^T X + \lambda D)^{-1} X^T y$$

where:

$$D = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

The zero in the top-left position means that the intercept is not penalised.

3.3 How Ridge Regression Stabilises the Model

Adding the ridge penalty improves the conditioning of the regression problem. The term λI or λD increases the diagonal entries of the matrix being inverted.

This is useful when predictors are highly correlated. Instead of allowing coefficients to become extremely large and unstable, ridge regression shrinks them toward zero.

This introduces some bias, but it can greatly reduce variance. In engineering prediction problems, this bias-variance trade-off often improves performance on new data.

3.4 Choosing the Regularisation Strength

The value of λ controls how strongly the coefficients are shrunk.

- If $\lambda = 0$, ridge regression reduces to ordinary least squares.
- If λ is small, ridge behaves similarly to OLS.
- If λ is large, coefficients are strongly shrunk toward zero.
- If λ is too large, the model may underfit.

The best value of λ is usually selected using validation data or cross-validation.

3.5 When Ridge Regression Works Best

Ridge regression is useful when:

- predictors are highly correlated,
- the model has many features,
- the number of features is close to or larger than the number of observations,
- OLS coefficients are unstable,
- prediction accuracy is more important than direct coefficient interpretation.

Ridge can be useful even when OLS is technically computable.

For example, ridge regression may be useful when modelling heat exchanger performance using many correlated temperature, pressure, and flow-rate measurements.

3.6 Drawbacks of Ridge Regression

3.6.1 No Automatic Feature Selection

Ridge shrinks coefficients toward zero, but it usually does not set them exactly to zero. Therefore, all features usually remain in the model.

If automatic feature selection is needed, methods such as Lasso regression may be more appropriate.

3.6.2 Feature Scaling Is Required

Ridge regression penalises coefficients according to their numerical size. Therefore, input features should usually be scaled before fitting the model.

For example, a model might use:

- pressure measured in Pascals,
- temperature measured in Kelvin,
- velocity measured in metres per second,
- heat flux measured in watts per square metre.

These variables naturally have different numerical scales. Without scaling, ridge may penalise some coefficients unfairly simply because of the units used.

A common approach is to standardise each feature:

$$z = \frac{x - \mu}{\sigma}$$

where:

- z is the standardised value,
- x is the original value,
- μ is the feature mean,
- σ is the feature standard deviation.

3.6.3 Coefficients Become Less Directly Interpretable

Because ridge deliberately shrinks coefficients, their numerical values are biased relative to OLS estimates.

This is often acceptable for prediction, but it should be remembered when interpreting coefficients physically.

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4 Model Assessment Metrics

After fitting a model, we need to quantify how well it performs.

No single metric tells the whole story. For engineering applications, it is often useful to report several metrics and inspect residual plots.

The three metrics covered here are:

1. **Mean Absolute Error**
2. **Root Mean Squared Error**
3. **Coefficient of Determination**

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5 Mean Absolute Error

The **Mean Absolute Error**, or **MAE**, is the average absolute difference between predictions and actual values.

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|$$

5.1 Interpretation

MAE gives the average magnitude of the prediction error in the same units as the target variable.

For example:

- if y is pressure drop in Pascals, MAE is in Pascals,
- if y is wall temperature in degrees Celsius, MAE is in degrees Celsius,
- if y is heat transfer coefficient in W m^{-2} K^{-1}, MAE is in W m^{-2} K^{-1}.

An MAE of 5 K means that, on average, the model predictions are 5 K away from the measured values.

5.2 Advantages of MAE

MAE is useful because:

- it is easy to interpret,
- it is in the same units as the target variable,
- it treats errors linearly,
- it is less sensitive to outliers than RMSE.

Because errors are not squared, an error of 10 units is penalised exactly twice as much as an error of 5 units.

5.3 Limitations of MAE

MAE has several limitations:

1. It does not heavily penalise large errors.
2. It may hide rare but dangerous prediction failures.
3. The absolute value function is not differentiable at zero.

The differentiability issue does not make MAE impossible to optimise, because subgradient-based methods can handle it. However, it can be less convenient than squared-error loss for some optimisation algorithms.

In thermofluid safety systems, MAE may be insufficient on its own because one very large pressure, temperature, or heat-flux error may be more important than many small errors.

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6 Root Mean Squared Error

The **Root Mean Squared Error**, or **RMSE**, is the square root of the average squared error.

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

6.1 Interpretation

RMSE is measured in the same units as the target variable.

For example:

- if y is pressure in Pascals, RMSE is in Pascals,
- if y is temperature in Kelvin, RMSE is in Kelvin,
- if y is heat flux in W m^{-2}, RMSE is in W m^{-2}.

Because RMSE squares errors before averaging, it gives greater weight to larger errors.

6.2 Advantages of RMSE

RMSE is useful because:

- it is in the same units as the target variable,
- it strongly penalises large errors,
- it is closely connected to least-squares regression,
- it is useful when large errors are especially undesirable.

In engineering applications, large prediction errors may correspond to unsafe or unacceptable designs. For example:

- underpredicting a wall temperature could lead to thermal failure,
- underpredicting pressure drop could lead to incorrect pump sizing,
- underpredicting heat flux could compromise cooling-system design.

RMSE is therefore useful when large errors should receive extra attention.

6.3 Limitations of RMSE

RMSE is highly sensitive to outliers.

If the dataset contains bad sensor readings, transient spikes, or failed simulation points, the squared term can make RMSE appear much worse than the model's typical performance.

For this reason, RMSE should often be reported alongside MAE and residual plots.

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7 Coefficient of Determination

The **Coefficient of Determination**, denoted R^2, measures the proportion of variation in the target variable that is explained by the model.

It is defined using two quantities:

1. Residual sum of squares, RSS
2. Total sum of squares, TSS

7.1 Residual Sum of Squares

The residual sum of squares is:

$$\mathrm{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

This measures the unexplained variation left over after fitting the model.

7.2 Total Sum of Squares

The total sum of squares is:

$$\mathrm{TSS} = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

where $\bar{y}$ is the mean of the observed target values.

TSS measures the total variation in the data relative to the mean.

A baseline model that predicts $\bar{y}$ for every observation has an error equal to TSS.

7.3 Formula for R Squared

The coefficient of determination is:

$$R^2 = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}}$$

7.4 Interpretation

If the model predicts perfectly, then:

$$\text{RSS} = 0$$

so:

$$R^2 = 1$$

If the model performs no better than predicting the mean, then:

$$\text{RSS} = \text{TSS}$$

so:

$$R^2 = 0$$

On test data, R^2 can be negative if the model performs worse than the mean-value baseline.

7.5 Advantages of R Squared

R^2 is useful because:

- it is dimensionless,
- it provides a quick measure of explained variation,
- it is familiar and widely reported,
- it can help compare models on similar datasets or the same target variable.

7.6 Limitations of R Squared

7.6.1 More Features Can Artificially Increase R Squared

For OLS fitted on the same dataset, R^2 never decreases when additional predictors are added, even if those predictors are random noise.

This can lead to false confidence in an overcomplicated model.

When comparing models with different numbers of predictors, use **Adjusted** R^2 or validation-set performance metrics.

Adjusted R^2 is:

$$R^2_{adj} = 1 - (1 - R^2)\frac{n-1}{n-p-1}$$

where:

- n is the number of observations,
- p is the number of predictors.

Adjusted R^2 penalises unnecessary predictors.

7.6.2 R Squared Does Not Prove the Model Structure Is Correct

A high R^2 does not guarantee that a linear model is physically appropriate.

For example, a nonlinear thermofluid relationship might produce a high R^2 over a limited operating range but fail outside that range.

Always inspect:

- residual plots,
- parity plots,
- error distributions,
- model behaviour across operating regimes.

7.6.3 R Squared Can Be Misleading Across Different Physical Problems

Because R^2 depends on the variability and noise level of the dataset, comparing it across very different physical systems can be misleading.

For example:

- in a highly turbulent flow dataset, an R^2 of 0.60 might be useful,
- in a simple laminar pipe-flow calibration, an R^2 of 0.95 might be poor if measurement uncertainty is very low.

Context matters.

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8 Practical Guidance for Thermofluid Engineers

8.1 When to Use OLS

Use OLS when:

- the relationship is approximately linear in the parameters,
- the number of predictors is modest,
- multicollinearity is not severe,
- interpretability is important,
- the dataset is reasonably clean.

OLS is a good first model and a useful baseline.

8.2 When to Use Ridge Regression

Use ridge regression when:

- predictors are correlated,
- the model has many features,
- OLS coefficients are unstable,
- predictive performance is more important than exact coefficient interpretation,
- the number of features is close to or greater than the number of observations.

Ridge is especially useful for sensor-rich thermofluid systems, where many measured variables are correlated.

8.3 Which Error Metric Should You Use?

Use **MAE** when you want a clear measure of typical error.

Use **RMSE** when large errors are especially important.

Use R^2 or **Adjusted** R^2 to understand explained variation, but do not treat them as proof that the model is physically correct.

A practical model report might include:

- MAE,
- RMSE,
- R^2,
- adjusted R^2 where appropriate,
- residual plots,
- parity plots,
- validation or test-set performance.

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9 Summary for the Engineer

- **OLS** is a simple and interpretable baseline model for approximately linear relationships.
- The OLS closed-form solution requires $X^T X$ to be invertible.
- Severe multicollinearity can make OLS coefficients unstable.
- **Ridge regression** adds an L2 penalty that shrinks coefficients and improves numerical stability.
- Ridge usually does not penalise the intercept.
- Ridge requires feature scaling when predictors have different units or numerical ranges.
- **MAE** measures typical absolute error and is less sensitive to outliers than RMSE.
- **RMSE** penalises large errors and is useful when large deviations are dangerous.
- R^2 measures explained variation, but it can be misleading if used without residual analysis.
- **Adjusted** R^2 is better than ordinary R^2 when comparing models with different numbers of predictors.
- Always combine statistical metrics with engineering judgement and knowledge of the operating regime.

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10 Recommended Workflow

For a thermofluid regression problem:

1. Inspect the raw data.
2. Check units, sensor quality, and obvious outliers.

3. Plot the target variable against key predictors.
4. Split the data into training and test sets.
5. Fit an OLS baseline model.
6. Check residuals and model assumptions.
7. Assess multicollinearity.
8. If needed, scale the features and fit ridge regression.
9. Choose the ridge penalty using cross-validation.
10. Compare models using MAE, RMSE, R^2, and adjusted R^2.
11. Inspect residual and parity plots.
12. Confirm whether the model behaves sensibly across relevant thermofluid regimes.

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11 Key Takeaway

Regression models are empirical tools. They can be extremely useful for thermofluid analysis, but they should not be treated as replacements for physics.

A good regression model should be:

- statistically sound,
- numerically stable,
- validated on unseen data,
- consistent with thermofluid understanding,
- interpreted within the operating range of the data.